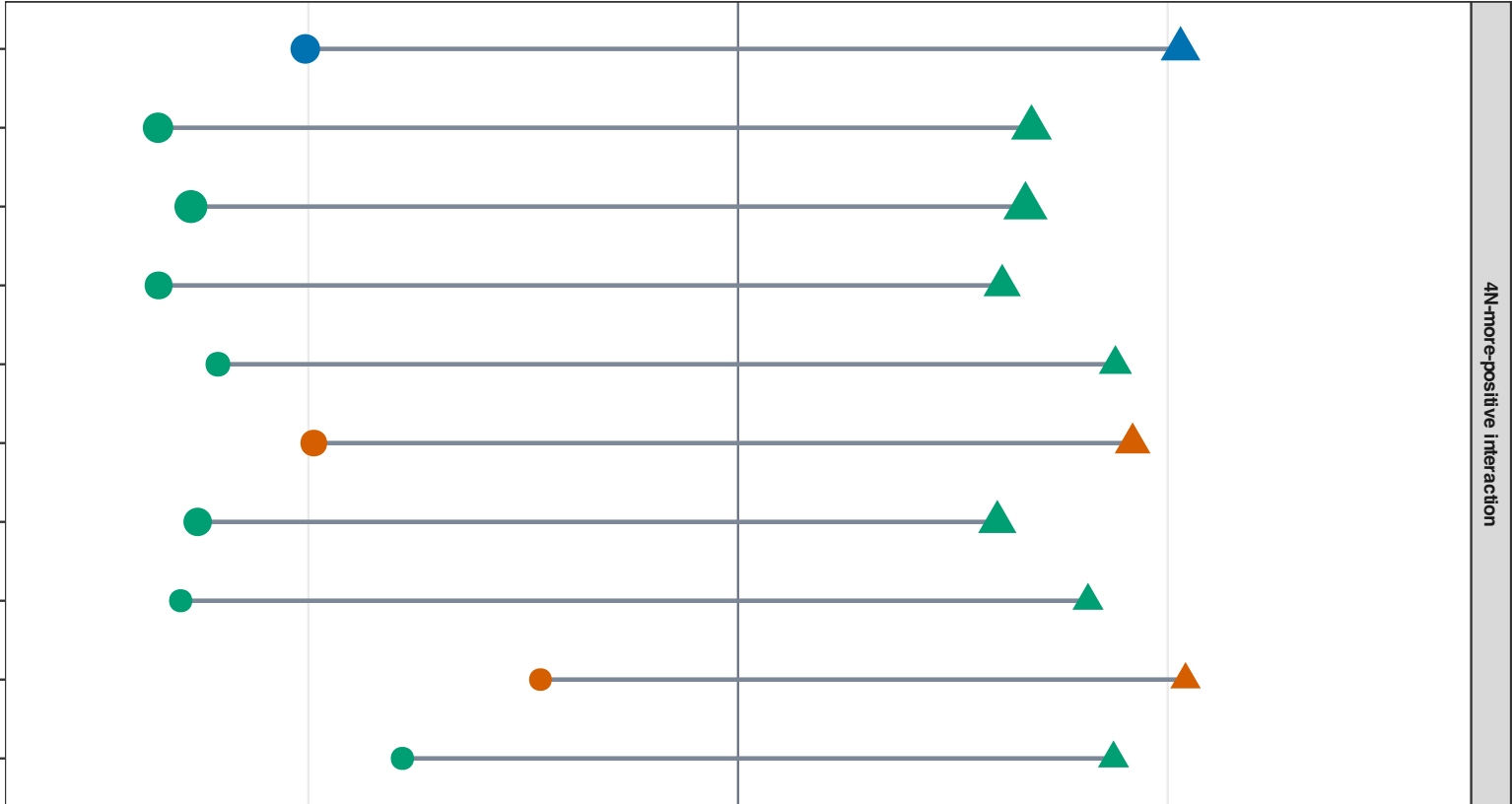
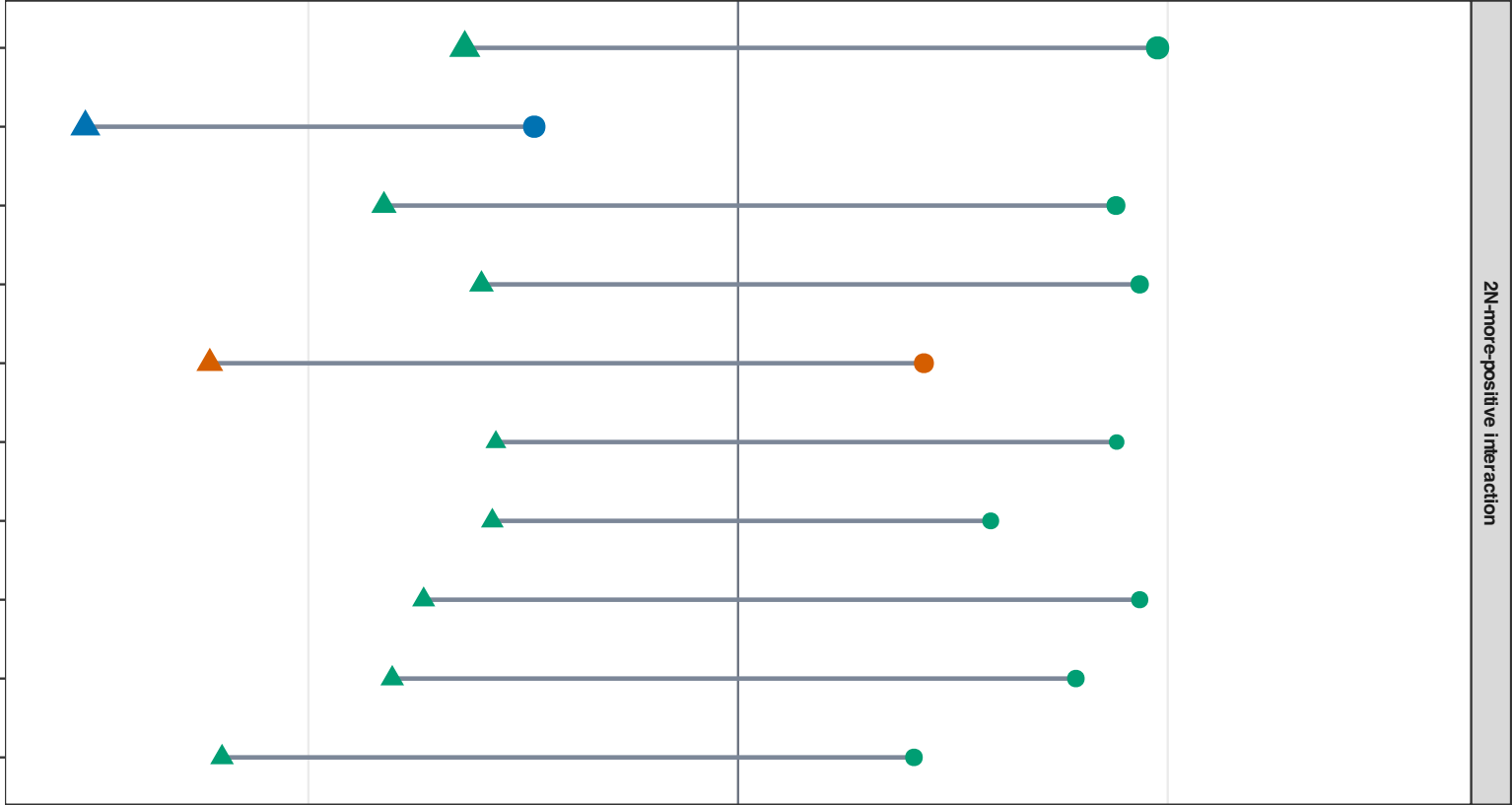


Origin-dependent treatment responses

Equal-dose treatment contrast; 0.296-0.486 pseudotime
2N: 242 cells across 8 mice; 4N: 175 cells across 8 mice

Collection • Hallmark • Reactome • GO BP Origin • 2N ▲ 4N $-\log_{10}$ FDR ● 10 ● 20



Pathways were selected as the top 10 negative and top 10 positive formal interaction NES across Hallmark, Reactome, and GO BP. Points show joint-model origin-specific treatment NES; connectors join the 2N and 4N estimates for each pathway. The upper panel has a more positive response in 2N and the lower panel in 4N; more positive can also mean less depleted. Point size represents formal-interaction FDR; all displayed pathways pass collection-wise BH FDR ≤ 0.05 . GRCh38 human-tumor mouse pseudobulks; adjusted for mean within-interval